

Stabilisation Strategies for Continuous Finite Element Simulation of Compressible Flows

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Continuum-Kinetic Models in Hypersonic Transitional Flows

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Brief history of CG-FEM for conservation laws

1982 **Hughes et al.**

Streamline-upwind/Petrov–Galerkin methods and their extension to systems of conservation laws, including the compressible Euler equations.

1990 **Johnson–Szepessy**

Convergence analysis for shock-capturing streamline-diffusion methods using residual-based artificial viscosity and entropy inequalities.

'87- **Löhner/Kuzmin/Turek**

Finite element flux-corrected transport, combining a monotone low-order method with limited antidiffusive fluxes to obtain high-resolution, nonoscillatory approximations.

'11- **Guermond**

Entropy-viscosity and invariant-domain methods using residual-based dissipation and convex limiting to preserve positivity, physical bounds, and entropy properties.

'90- **Abgrall**

Residual-distribution methods that distribute element residuals among the degrees of freedom to construct multidimensional, locally conservative, and entropy-stable schemes.

Space–Time Streamline Diffusion/Shock Capturing

► Hyperbolic system of conservation laws

$$\partial_t u + \nabla \cdot f(u) = 0 \quad \text{in } Q = \Omega \times (0, T).$$

Define the space–time residual: $R(U_h) := U_{h,t} + \nabla \cdot f(U_h)$, and let V_h be a *space–time* finite element space.

► SD–SC formulation (Szepessy)

$$\sum_{K \subset Q} \left[\int_K R(U_h) \left(v_h + \underbrace{\delta_K a(U_h) \cdot \nabla_{x,t} v_h}_{\text{streamline-diffusion}} \right) dx dt + \int_K \underbrace{\varepsilon_K \nabla_{x,t} U_h \cdot \nabla_{x,t} v_h}_{\text{shock-capturing}} dx dt \right] =$$

for all $v_h \in V_h$, where $a(U_h) = (1, f'(U_h))$.

Stabilization parameters: $\delta_K \sim h_K$, $\varepsilon_K = C_{sc} h_K^t |R(U_h)|$, $t \in (1, 2)$.

► Entropy viscosity obtained by taking $\delta_K = 0$ and taking the residual of the entropy equation in ε_K . Can use any time discretisation.

Szepessy: Theoretical results for SD-SC CG-FEM I

► Discrete entropy inequality

The stabilised finite element solution satisfies an entropy estimate of the form

$$\int_{\Omega} \eta(u_h(T)) \, dx + \int_0^T D_h(u_h) \, dt \leq \int_{\Omega} \eta(u_h(0)) \, dx,$$

where $D_h(u_h) \geq 0$ represents the dissipation generated by streamline diffusion and shock capturing.

► Strong convergence implies convergence to entropy solutions for systems

If u_h converges strongly, consistency and the discrete entropy inequality imply that the limit is a weak solution satisfying the corresponding entropy inequality.

► Compactness leads to convergence for scalar conservation laws

Uniform bounds combined with compensated compactness yield a subsequence converging to the unique entropy solution. Shock-capturing viscosity supplies the maximum-norm control needed to complete the argument.

⁰C. Johnson and A. Szepessy, Math. Comp. 49 (1987), 427–444; C. Johnson, A. Szepessy and P. Hansbo, Math. Comp. 54 (1990), 107–129.

Why DG methods for conservation laws? I

For a mesh $\mathcal{T}_h$, the DG approximation is polynomial on each element but is allowed to be discontinuous across element interfaces:

$$V_h^k = \{v_h \in L^2(\Omega) : v_h|_K \in \mathbb{P}_k(K), \quad K \in \mathcal{T}_h\}.$$

► Method-of-lines formulation

The spatial discretisation produces a system of ordinary differential equations

$$M \frac{dU}{dt} = R(U),$$

which can be combined with explicit Runge–Kutta time-stepping methods.

► Systematic high-order approximation

Higher spatial order is obtained by increasing the local polynomial degree k .

Why DG methods for conservation laws? II

► **Local conservation**

Using a single-valued numerical flux on every interface gives an exact balance on each element:

$$\frac{d}{dt} \int_K u_h dx + \int_{\partial K} \hat{F}(u_h^-, u_h^+, n_K) ds = 0.$$

► **Upwinding through numerical fluxes (c.f. Streamline diffusion)**

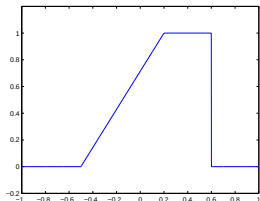
Riemann solvers and monotone numerical fluxes introduce dissipation in the characteristic directions and provide a natural treatment of inflow, outflow, and nonlinear wave propagation.

► **Nonlinear limiting near shocks (c.f. Shock capturing)**

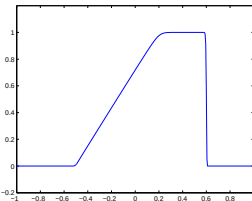
TVD, ENO, WENO, moment, and bound-preserving limiters can be applied locally to troubled elements while leaving the high-order polynomial unchanged in smooth regions.

Examples, stabilised versus unstabilised FEM

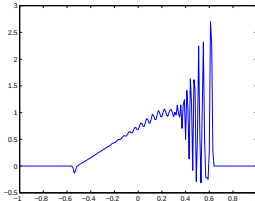
- Example: Burgers' equation: $\partial_t u + \frac{1}{2} \partial_x u^2 = 0$



Exact solution;



stabilised soln;



energy conserving soln.

Problems with established methods:

- Discontinuous Galerkin: large number of DOFs (factor 6 in 2D, factor 20 in 3D).
- SUPG: stability issues for complex problems.

$$(\partial_t u_h + \beta \cdot \nabla u_h + \omega(u_h), v_h + \delta \beta \cdot \nabla v_h)_\Omega = (f, v_h + \delta \beta \cdot \nabla v_h)_\Omega$$

The Interior Penalty Method, 1976

Interior Penalty Procedures for Elliptic and Parabolic Galerkin Methods

Jim Douglas, Jr., and Todd Dupont

University of Chicago, Chicago, Illinois 60637

Abstract. Standard Galerkin methods based on C^0 -piecewise-polynomial spaces often can lead to unsatisfactory approximations of solutions of problems having dominant transport terms. A penalty on the jump in the normal derivative across the interior edges of elements can produce an apparent stiffness intermediate between C^0 and C^1 , and such a method is proposed and analyzed for elliptic and parabolic equations.

- Douglas, J., Jr., Dupont, T. Interior penalty procedures for elliptic and parabolic Galerkin methods. In: *Computing Methods in Applied Sciences* (Versailles, 1975), Lecture Notes in Physics 58, Springer, 1976, pp. 207-216.

Interior Penalty Method: Core Concept I

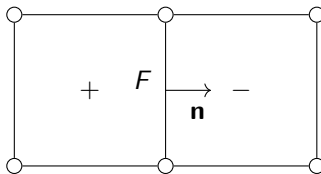
- ▶ For a triangulation $\mathcal{T}_h$ with interior faces F , define the **jump of the normal derivative**:

$$\left[\frac{\partial v}{\partial n} \right] = \frac{\partial v^+}{\partial n} - \frac{\partial v^-}{\partial n} \quad \text{on } F.$$

- ▶ Consider a penalty term of the form

$$S(v, w) = \sum_F \int_F \sigma_F \left[\frac{\partial v}{\partial n} \right] \left[\frac{\partial w}{\partial n} \right] ds,$$

where $\sigma_k \geq 0$ is a weight parameter.



Interior Penalty Method: Core Concept II

- ▶ Let V_h denote the standard space of continuous finite element functions of degree $\leq k$, mesh-size h .
- ▶ The full bilinear form becomes

$$A(v, w) = B(v, w) + S(v, w),$$

where

$$B(v, w) := \int_{\Omega} \varepsilon \nabla v \cdot \nabla w + (\mathbf{b} \cdot \nabla w + r v) w$$

Formulation: Find $u_h \in V_h$ such that

$$A(u_h, v_h) = (f, v_h)_{\Omega}, \quad \forall v_h \in V_h.$$

- ▶ Smooth $u \implies J(u, v_h) = 0$, **the method is consistent.**
- ▶ Douglas & Dupont did not prove stability in the hyperbolic limit and the method was forgotten!

The revival: 2004 Burman and Hansbo¹

- ▶ Burman & Hansbo (2004) revisit and generalize this idea for convection-diffusion-reaction problems.
 - ▶ **least-squares penalty** on gradient jumps
 - ▶ handles the **hyperbolic** (vanishing-diffusion) limit
 - ▶ introduces **additional nonlinear stabilisation** to enforce monotonicity
- ▶ Similar results as Streamline-Diffusion or discontinuous Galerkin.
- ▶ Key observation, when $\sigma = |\mathbf{b}|h^2$:

$$\inf_{w_h \in V_h} \|h^{1/2}(\mathbf{b} \cdot \nabla v_h - w_h)\|_{\Omega} \lesssim S(v_h, v_h)^{1/2}.$$

- ▶ Conceptually: a higher-order, continuous analogue of DG upwind schemes.
- ▶ Mesh-dependent triple norm with control of the streamline derivative.

$$\|v_h\| := \|\varepsilon^{1/2} \nabla v_h\|_{\Omega} + \|h^{1/2} \mathbf{b} \cdot \nabla v_h\|_{\Omega} + \|r^{1/2} v_h\|_{\Omega} + S(v_h, v_h)^{1/2}$$

- ▶ A priori estimate ($k = 1$):

$$\|u - u_h\| \leq C(\varepsilon^{1/2} h + |\mathbf{b}|h^{3/2} + r^{1/2} h^2) \|u\|_{H^2}.$$

¹EB, P. Hansbo, Comput. Methods Appl. Mech. Engrg. 193 (2004), no. 15-16, 1437-1453.

Friedrichs Systems - linear model for conservation laws²

- ▶ A Friedrichs system is a first-order system of PDEs, $u : \Omega \rightarrow \mathbb{R}^m$

$$\mathbb{T}u = Ku + \sum_{k=1}^d A_k \partial_k u = f,$$

where $K \in L^\infty(\Omega)^{m \times m}$, $A_k \in W^{1,\infty}(\Omega)^{m \times m}$ satisfy:

- ▶ $A_j = A_j^\top$ (symmetry),
 - ▶ $K + K^\top - \sum_{j=1}^d \partial_k A_j \geq 2\mu_0 I_m$ (positivity), $\mu_0 > 0$.
- ▶ Boundary conditions are described via a **boundary operator**

$$D = \sum_{j=1}^d n_j A_j,$$

and an operator M splitting $\partial\Omega$ into "incoming/outgoing" parts.

- ▶ Many PDEs fit this framework: **symmetric hyperbolic systems**, advection-reaction, advection-diffusion in mixed form, linear elasticity, Maxwell in elliptic regime.

²EB, A. Ern, Math. Model. Numer. Anal. 41 (2007), no. 1, 55–76.

Continuous Interior Penalty for Friedrichs Systems

- ▶ Goal: design a stable CFEM scheme for $\mathbb{T}u = f$ using the IP method.
- ▶ Discrete space: $W_h = [V_h]^m$, V_h continuous k th-order FE space.
- ▶ Stabilisation: add penalties on **jumps of normal derivatives**

$$S(v_h, w_h) = \sum_{F \in \mathcal{F}_i} (S_F [\![\nabla v_h]\!], [\![\nabla w_h]\!])_F,$$

with S_F chosen so that

$$h_F^2 |D_F| \lesssim S_F \lesssim h_F^2 I_m, \quad D_F = \sum_{k=1}^d A_k n_{F,k}.$$

- ▶ Weak enforcement of **boundary conditions** through

$$\frac{1}{2}((M_F - D_F)v_h, w_h)_F, \quad F \subset \partial\Omega,$$

mirroring the DG boundary fluxes. M_F boundary "upwind" penalty.

- ▶ Discrete variational problem: find $u_h \in W_h$ such that $\forall y_h \in W_h$,

$A_h(u_h, y_h) = (f, y_h)_\Omega, \quad A_h(\cdot, \cdot) = (\mathbb{T}\cdot, \cdot)_\Omega + S(\cdot, \cdot) + \text{boundary terms.}$

Stability and Convergence for Friedrichs Systems

- ▶ The CIP bilinear form $A_h(\cdot, \cdot)$ is L^2 -**coercive**:

$$A_h(v_h, v_h) \gtrsim \|v_h\|_{L^2(\Omega)}^2 + \sum_{F \in \mathcal{F}_\partial} \|v_h\|_{M,F}^2 + \sum_{F \in \mathcal{F}_i} h_F^2 \|[\![\nabla v_h]\!]\|_F^2.$$

- ▶ Inf-sup stability:

$$\|v_h\| \lesssim \sup_{w_h \neq 0} \frac{A_h(v_h, w_h)}{\|w_h\|},$$

where $\|\cdot\|$ includes the h weighted graph norm $\|h^{1/2}\mathbb{T}v_h\|_\Omega$.

- ▶ Consistency: if $u \in H^2(\Omega)^m$, then

$$A_h(u - u_h, y_h) = 0 \quad \forall y_h \in W_h.$$

- ▶ Convergence result:

$$\|u - u_h\| \lesssim \inf_{v_h \in W_h} \|u - v_h\|_*,$$

so that for $u \in H^{k+1}$,

$$\|u - u_h\| \lesssim h^{k+\frac{1}{2}} \|u\|_{H^{k+1}}.$$

- ▶ Same philosophy as CIP for scalar transport, but extended to the fully coupled matrix-valued operator $\mathbb{T}$.

Example, Maxwell's equations as a first order system

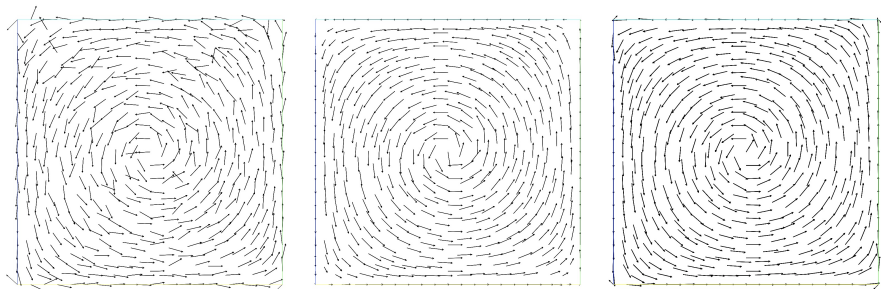


FIGURE 4. Maxwell's equations in the elliptic regime: approximate solution (H_1, H_2) obtained with CFEM without stabilization (left), with stabilization on H and E (center), and with stabilization on E only (right).

Runge–Kutta Time Integration for Friedrichs' Systems³

We consider the evolution problem

$$\partial_t u + Au = f, \quad (M - D)u = 0 \text{ on } \partial\Omega$$

where $A = \sum_{j=1}^d A_j \partial_j$ is a Friedrichs operator (symmetric A_j).
Spatial semidiscretization by stabilised FEM (CIP or DG) yields

$$\frac{d}{dt} u_h + A_h u_h = f_h, \text{ where } (A_h v_h, w_h)_\Omega = A_h(v_h, w_h)$$

- ▶ **Time discretisation by RK2 or RK3.**
- ▶ Analysis leads to a CFL constraint: $\tau \leq C h/\lambda$, where λ is max wave speed.
- ▶ Error estimate for smooth solution:

$$\|u(T) - u_h(T)\|_L \lesssim \tau^q + h^{k+1/2} \quad (\text{RK}q, q=2,3)$$

³EB, A. Ern, M. Fernández, SIAM J. Numer. Anal. 48 (2010), no. 6, 2019-2042.

Compressible Euler Equations: IP-FEM

Unknown: $u = (\rho, m, E)$ in V_h^{d+2} (mass, momentum, total energy)

Weak semi-discrete FE formulation

Find $u_h(t) \in V_h^{d+2}$ such that for all $w_h \in V_h^{d+2}$,

$$\underbrace{(\partial_t u_h, w_h)_\Omega - \sum_{i=1}^d (f_i(u_h), \partial_{x_i} w_h)_\Omega}_{\text{standard Galerkin CGFEM}} + \underbrace{s_0(u_h; u_h, w_h) + s_1(u_h; u_h, w_h)}_{\text{stabilisation terms}} = 0.$$

Fluxes (Euler system)

$$f_1(u) = \begin{pmatrix} m_1 \\ m_1 v_1 + p \\ m_2 v_1 \\ (E + p)v_1 \end{pmatrix}, \quad f_2(u) = \begin{pmatrix} m_2 \\ m_1 v_2 \\ m_2 v_2 + p \\ (E + p)v_2 \end{pmatrix}, \quad p = (\gamma - 1) \left(E - \frac{|m|^2}{2\rho} \right).$$

Compressible Euler: Linear & Nonlinear Stabilisation⁴

Two stabilisation mechanisms

- ▶ **Linear symmetric stabilisation:** Gradient Jump Penalty (GJP / CIP)

$$s_0(u_h; v_h, w_h) = \sigma_0 \sum_K \int_{\partial K} h_K^2 (1 - \varpi_K(u_h)) |A_n(u_h)| \llbracket \nabla v_h \cdot n \rrbracket \llbracket \nabla w_h \cdot n \rrbracket.$$

Controls high-frequency pollution, guarantees localisation of error.

- ▶ **Nonlinear stabilisation:** Shock-capturing/Entropy-viscosity term:

$$s_1(u_h; v_h, w_h) = \sigma_1 \sum_K \varpi_K(u_h) h_K \|A(u_h)\|_{\infty, K} (\nabla v_h, \nabla w_h)_K.$$

Adds sufficient dissipation where entropy condition violated.

- ▶ **Switching function** $\varpi(u_h) \in [0, 1]$: Detects shock zones/entropy violation.

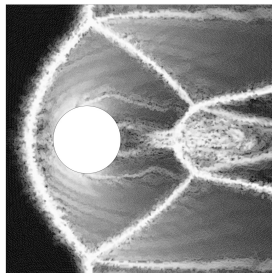
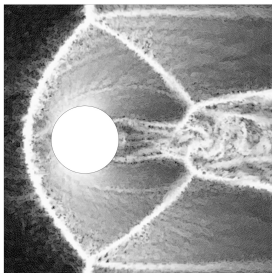
Objective: Preserve invariants (s_1) and achieve $O(h^{k+1/2})$ in smooth regions (s_0).

⁴EB, SIAM J. Sci. Comput. 45 (2023), no. 1, A96-A122.

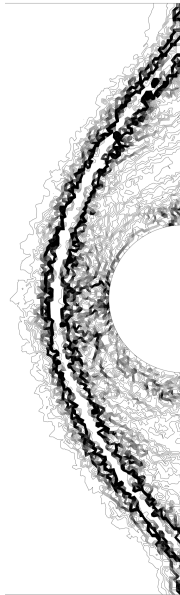
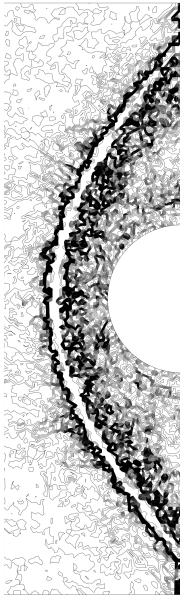
Insufficiency of Nonlinear Stabilization Alone I

Observation: Nonlinear artificial viscosity alone does not prevent the propagation of spurious gradient oscillations into smooth regions of the solution.

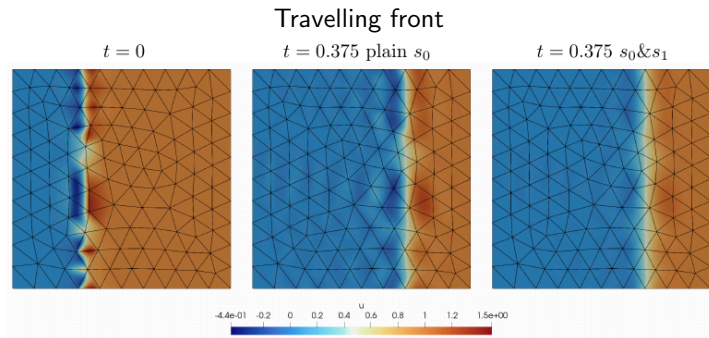
- ▶ **Localization principle fails:** oscillations generated near shocks do not remain localized.
- ▶ Nonlinear stabilization alone (**left below**) leads to (i) gradient oscillations upstream of the shock and (ii) "grittiness" caused by unresolved oscillations.
- ▶ addition of j_0 recovers localization (**right below**)



Insufficiency of Nonlinear Stabilization Alone II



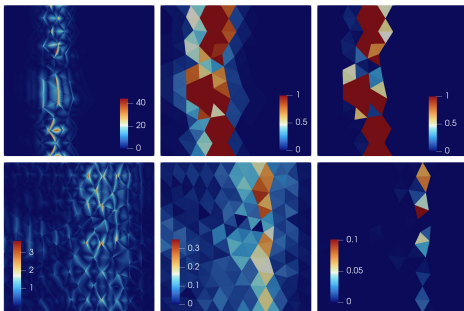
Localisation for linear and nonlinear stabilisation



- ▶ The stabilisation s_0 reduces dispersion errors, may violate the entropy condition close to shocks.
- ▶ Nonlinear artificial diffusion s_1 controls shocks, but gradient oscillations may enter smooth regions.
- ▶ Aim: prove local $O(h^{k+1/2})$ accuracy where the exact solution is smooth for

$$\mathcal{L}u := \partial_t u + \beta \cdot \nabla u = f, \quad \beta \in \mathbb{R}^d$$

Residual switch between s_0 and s_1

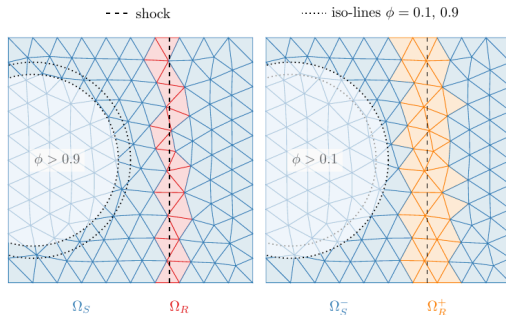


- Plot of: (left) $|\mathcal{L}u_h - f|$; (middle) ϖ_K , $\alpha = 1$; (right) ϖ_K , $\alpha = 4$.
- Top row, $T = 0.05$, bottom row $T = 0.375$.
- For $K \in \mathcal{T}_h$, define: $\varpi_K(w_h) = \min \left\{ 1, \left(\frac{h_K R_K(w_h)}{U} \right)^\alpha \right\}$, where

$$R_K(w_h) = \rho_1 \| [\nabla w_h \cdot n] \|_{\infty, \partial K} + \rho_2 \| \mathcal{L}w_h - f_h \|_{\infty, K},$$

- $(1 - \varpi_K)$ weights s_0 ; ϖ_K weights the artificial diffusion term s_1 .
- Large α confines diffusion more tightly to the shock.

Weighted localisation along characteristics



Choose a positive weight transported by the flow (follows the characteristics),

$$(\partial_t + \beta \cdot \nabla)\phi = 0, \quad \phi(x, t) = \psi(x - x_0 - \beta t),$$

with transition width $K\sqrt{h}$ and

$$|D^\ell \phi| \lesssim (K\sqrt{h})^{-\ell} \phi, \quad |\nabla \phi| \lesssim K^{-1} h^{-1/2} \phi.$$

- $\phi = 1$ in the smooth observation region Ω_S and small in the rough region $\Omega_R(t)$.

Weighted stability

To prove local stability choose the test function

$$w_h = \pi_h [\phi^2 (v_h + \theta h i_{\text{av}}(\partial_t v_h + \beta \cdot \nabla v_h))] .$$

For $h^{1/2}/K$ sufficiently small and $\sigma_1 \gtrsim \sigma_0$,

$$\|v_h(T)\|_\phi^2 + C_\theta \int_0^T \|v_h\|_{y_h, S, \phi}^2 dt \lesssim K^{-2} \int_0^T \|v_h\|_\phi^2 dt + \|v_h(0)\|_\phi^2 + 2 \int_0^T (\mathcal{L}v_h, w_h) dt .$$

The weighted norm is defined by

$$\|v_h(T)\|_\phi = \|v_h(T)\phi(T)\|_{L^2(\Omega)}$$

The stronger weighted norm $\|v_h\|_{y_h, S, \phi}$ controls the ϕ weighted

- ▶ material residual $h^{1/2}(\partial_t v_h + \beta \cdot \nabla v_h)$,
- ▶ CIP gradient jumps,
- ▶ switched artificial diffusion.

Local error estimate.

Let $e_h = u_h - \pi_h u$. For $\alpha \in [1, 2)$,

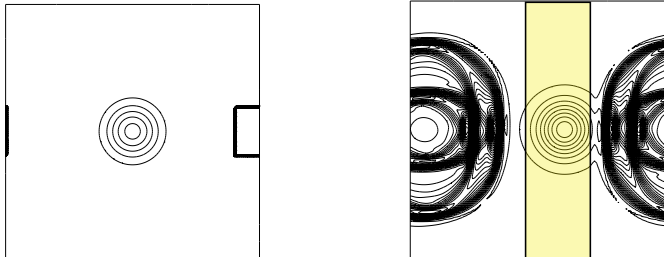
$$\|e_h(T)\|_\phi^2 + C_\theta \int_0^T \|e_h\|_{u_h, S, \phi}^2 dt \lesssim h^{2k+1} \mathcal{R}(u) + C(1 - \alpha/2) h^{\frac{2+\alpha}{2-\alpha}} \int_0^T \|\nabla u\|_{L^\infty(\Omega_S)}^{\frac{4}{2-\alpha}} dt,$$

where

$$\mathcal{R}(u) := \int_0^T \left(\underbrace{\|D^{k+1}u\|_{\Omega_S(t), \phi}^2}_{\text{local in smooth region}} + \underbrace{\|\beta \cdot \nabla u\|_{L^1(\Omega)}^2 + \|u\|_{L^2(\Omega)}^2}_{\text{global low order norms}} \right) dt.$$

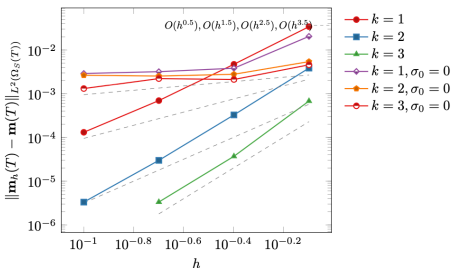
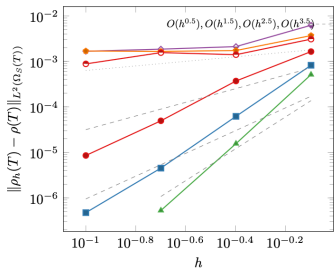
- ▶ Local L^2 accuracy: $\|e_h(T)\|_{\Omega_S} = O(h^{k+1/2})$.
- ▶ $H^{k+1} \cap W^{1,\infty}$ regularity is required only in $\Omega_S(t)$.
- ▶ For $\alpha \geq 2$ the bound holds without the last term when $h\|\nabla u\|_{L^\infty(\Omega_S)}$ is sufficiently small.

Laminar vortex and shock interaction



- ▶ Left, $T = 0$, right $T = 2$.
- ▶ We compute the L^2 -error in the yellow zone, $\Omega_S(T = 2) = [5, 6.5] \times [0, 10]$.
- ▶ Timediscretisation using Explicit Runge-Kutta of order 3.

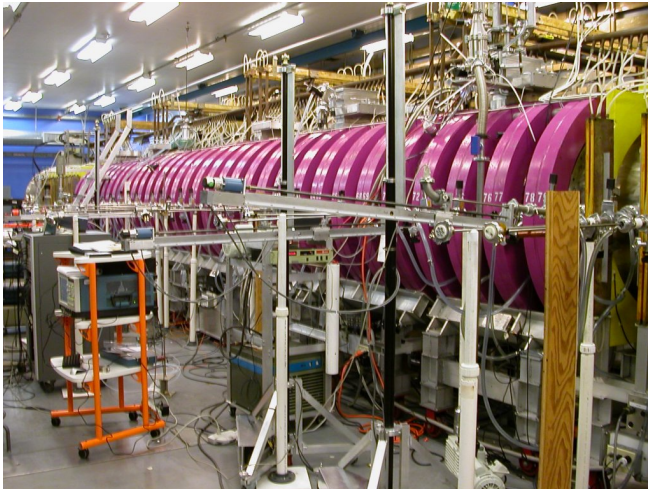
Laminar vortex and shock interaction, convergence in Ω_S



Observations

- ▶ $\sigma_1 = 0.5$, $\sigma_0 \in \{0, 0.01\}$, $U = 0.5$.
- ▶ For $k = 1, 2, 3$, $\sigma_0 = 0.01$, $\sigma_1 = 0.5$, both variables converge at least as $\mathcal{O}\left(h^{k+\frac{1}{2}}\right)$.
- ▶ For $k = 1, 2, 3$, $\sigma_0 = 0$, $\sigma_1 = 0.5$, both variables converge at $\mathcal{O}\left(h^{\frac{1}{2}}\right)$.
- ▶ Validates the local error analysis in the nonlinear case, for $\sigma_0 = 0.01$, $\sigma_1 = 0.5$.
- ▶ high order convergence fails in the smooth zone when $\sigma_0 = 0$.

The Large Plasma Device (LAPD)



"Edge turbulence is a key to the viability of fusion reactors because it is the main mechanism transporting heat and particles across magnetic field lines in the boundary region." - ChatGPT

Reduced Braginskii Model for the LAPD

$$\frac{dn}{dt} = \frac{\partial(nV_e)}{\partial z} + F_n, \quad (1)$$

$$\frac{dT_e}{dt} = \frac{2}{3} T_e \left[\frac{e}{n} 0.71 \frac{\partial j_{\parallel}}{\partial z} - \frac{2}{3} \frac{\partial V_e}{\partial z} - V_e \frac{\partial T_e}{\partial z} \right] + F_T, \quad (2)$$

$$\frac{d\nabla_{\perp}^2 \phi}{dt} = V_i \frac{\partial(\nabla_{\perp}^2 \phi)}{\partial z} + \frac{m_i}{2\Omega_i e} \frac{1}{n} \frac{\partial j_{\parallel}}{\partial z}, \quad (3)$$

$$\frac{dV_i}{dt} = V_i \frac{\partial V_i}{\partial z} - \frac{1}{n} \frac{\partial p_e}{\partial z}, \quad (4)$$

$$m_e \frac{dV_e}{dt} = m_e V_e \frac{\partial V_e}{\partial z} - \frac{T_e}{n} \frac{\partial n}{\partial z} \boxed{-e \frac{\partial \phi}{\partial z} + 1.71 \frac{\partial T_e}{\partial z}} + e \nu j_{\parallel}. \quad (5)$$

$$p_e = n T_e, \quad j_{\parallel} = en(V_i - V_e), \quad \frac{df}{dt} = \frac{\partial f}{\partial t} - \frac{c}{B} [\phi, f].$$

The Bohm sheath conditions at trans sonic outflow (!)

$$\left. \begin{aligned} V_i &= c_s \quad (\sim \sqrt{T_e/m_i}) \\ V_e &= c_s \exp(\Lambda - \frac{\phi}{T_e}) \end{aligned} \right\} \text{ necessary for stability when red terms included.}$$

2D Reduced Braginskii Model

2D reduced equations obtained by integrating along z :

$$\frac{dn}{dt} = \frac{nc_s}{R} \exp\left(\Lambda - \frac{e\phi}{T_e}\right) + F_n, \quad (6)$$

$$\frac{d}{dt} \nabla_{\perp}^2 \phi = \frac{c_s}{m_i 2 \Omega_i e R} \left[1 - \exp\left(\Lambda - \frac{e\phi}{T_e}\right) \right], \quad (7)$$

$$\frac{dT_e}{dt} = \frac{2}{3} \frac{T_e c_s}{R} \left(1.71 \exp\left(\Lambda - \frac{e\phi}{T_e}\right) - 0.71 \right) + F_T. \quad (8)$$

- **Sheath coupling** appears via the $\exp(\Lambda - \phi/T_e)$ terms, **now in the bulk**.
- **The Galerkin semi-discretisation in space** can be written: find $U_h(t) \in \mathcal{W}_h$ such that, for all $t > 0$, $\forall W_h \in \mathcal{W}_h$,

$$(\partial_t U_h + \sum_{i=1}^d A_i(U_h) \partial_i U_h + B(U_h), W_h)_{\Omega} + S(U_h, W_h) = (F, W_h)_{\Omega}.$$

 (9)

3D computation of transition, simplified Bohm conditions,
 $T = \phi/\Lambda$ in red term

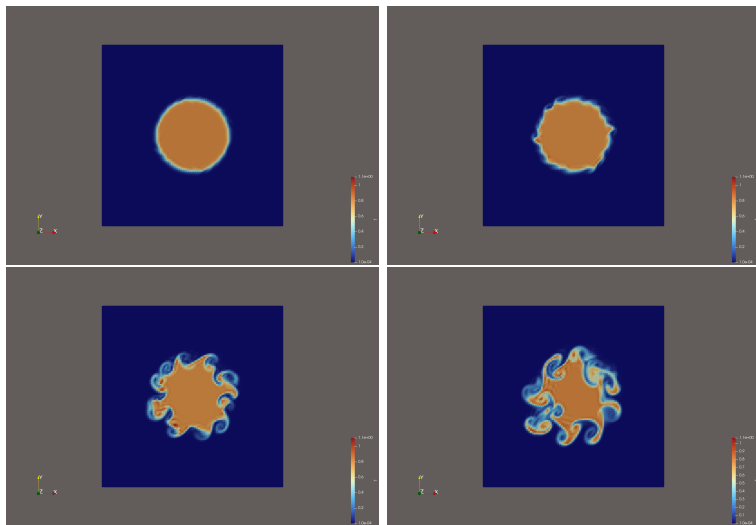


Figure: Temperature cross section using $\tau_1 = 5 \cdot 10^{-4}$ on a mesh, $100 \times 100 \times 8$, cross section at middle of tube for times $t = 0.256, 0.51, 0.96, 1.54$.

Conclusions

- ▶ The interior penalty method has similar stability properties as discontinuous Galerkin methods
- ▶ Accuracy in the smooth zone requires a weighted combination of different stabilising terms
- ▶ Convergence analysis similar to that of Szepessy for SUPG in progress
- ▶ Imposition of the Bohm sheath conditions in progress